

# AMRANE ANES

AI Engineer | Machine Learning | Full-Stack Developer

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## SUMMARY

AI Engineering graduate specialising in machine learning, computer vision, nlp and graph neural networks. Built and deployed an end-to-end multimodal AI pipeline for solar panel defect detection and maintenance recommendation as a graduation project, covering deep learning, model fine-tuning, knowledge graph construction, and RGAT-based classification. Complementary full-stack development background with production web and NLP projects. Seeking an AI/ML or software engineering role with an international or remote-friendly team.

## EDUCATION

### Master's in Intelligent Systems Engineering (AI)

September 2024 – June 2026 | Algeria

Université Saad Dahleb Blida

Graduation project: AI-Based Predictive Maintenance for Solar Panels Using Multimodal Knowledge Graph.

Implemented EfficientNet-B0 fine-tuning, preprocessing (CLAHE), MMKG construction linking visual features and structured metadata, and Relational Graph Attention Network (RGAT) training for fault classification.

Core coursework: Deep Learning, Graph Neural Networks, Computer Vision, Intelligent Agents & Multi-Agent Systems.

### Bachelor's in Computer Science

Graduated 2024 | Algeria

Université Saad Dahleb Blida

Final-year project: Tabibi full-stack web platform for online medical appointment booking.

## TECHNICAL SKILLS

**AI / ML:** PyTorch, PyTorch Geometric, EfficientNet, RGAT, Scikit-learn, spaCy, OpenCV, NumPy, Pandas

**Knowledge Graphs & GNNs:** MMKG construction, graph attention networks (GAT / RGAT), property graphs

**NLP:** Text classification, sentiment analysis, abusive speech detection & rewriting, Arabic NLP

**Programming Languages:** Python, JavaScript, TypeScript

**Web & Backend:** Next.js, React, Node.js, Express.js, MySQL, Tailwind CSS, REST APIs

**Tools & DevOps:** Git, GitHub, Docker (basics), Vercel, n8n (workflow automation), VS Code

## PROJECTS

### AI-Based Predictive Maintenance for Solar Panels Using MMKG

Master's Graduation Project | Jun 2026

Built an end-to-end multimodal pipeline (EL image preprocessing, EfficientNet-B0 fine-tuning, N-MMKG construction in PyTorch Geometric, RGAT training) for 5-class solar cell defect classification across 15,900 cells / 265 panels.

Achieved F1-Macro of 0.8092, outperforming RGCN and RGAT+MLP baselines; ablation study showed visual (EL image) modality is the primary discriminative signal (F1 drops to 0.21 without it).

Deployed as an interactive Streamlit inference dashboard; incubated as a startup (SolarGuard) at CDE Blida 1 under Ministerial Decree 1275.

### AI News Automation Agent

Personal Project | 2026

Built a reactive agent pipeline in n8n: scrapes Hacker News AI threads, processes content via Groq LLM, and publishes French-language summaries to Discord fully automated.

Framed and documented the system using the Wooldridge & Jennings agent-property framework (reactivity, proactivity, social ability).

### Real-Time Fire and Smoke Detection

University Project | 2026

Built a YOLOv8-based real-time fire and smoke detection system, achieving mAP@0.5 of 0.803.

Applied advanced image data augmentation, validation dataset cleaning, explainable AI techniques, and model comparison experiments.

### Abusive-to-Friendly Speech Transformer

NLP Project | 2025

Developed an NLP pipeline to detect hostile or abusive text and rewrite it in a neutral tone; supports Arabic and English.

Includes a CLI sentiment analysis tool for both languages.

PDF-Based AI Chatbot (RAG)

Personal Project | 2025

• Built a retrieval-augmented generation chatbot that ingests PDF documents and answers natural-language questions via an NLP pipeline (chunking, embedding, retrieval).

Tabibi Medical Appointment Platform

Bachelor's Final-Year Project | Juin 2024

• Full-stack web application: Node.js / Express REST API, MySQL database, React / Tailwind CSS frontend.

• Features: doctor scheduling, patient records, appointment booking and management.

LANGUAGES

• Arabic: Native

• English: B2 • Professional working proficiency

• French: B1 • Intermediate

ADDITIONAL

• Open to relocation and fully remote positions; actively exploring international AI engineering opportunities.

• Consistent self-learner: built multiple production-quality projects independently alongside full-time academic work.

• Interests: generative AI, agent-based systems, graph neural networks, startup culture, machine learning and computer vision and NLP.